

## Bike to Work 2024: CERN achieves the best participation rate

Despite the unstable weather in May and June, 1037 staff and users took up the challenge and joined one of the 276 CERN teams, putting the Organization in first place for the participation rate in its category



On 19 June 2024, the CERN Critical Mass event brought together many colleagues who enjoyed a bicycle ride around the Meyrin site before posing for a group photo in front of the (currently empty) Main Building. (Image: CERN)

This year's [Bike to Work](#) campaign has come to an end and the outcome is very positive for CERN, with a higher rate of participation than ever before: 1037 people took up the challenge and joined one of the 276 CERN teams. Congratulations!

Bike to Work encourages workers in companies all over Switzerland to commute to work by bicycle as often as possible throughout May and June. This year, a total of 28% of CERN's 3700 eligible population (staff and users) signed up to take part, placing the Organization in first place for the participation rate in its category (companies with 1000-4999 employees).

Unfortunately, the CERN participants did not have much luck in winning [team prizes](#) this year, but some entrants did win [individual prizes](#) – well done to them!

[Road safety](#) is a key consideration in making sure that CERN cyclists can ride safely around the site alongside all other road users. Road incident and accident statistics are monitored by the HSE unit and reported to the SCE department, which works tirelessly to [continuously improve the cycling and pedestrian routes both inside and outside CERN](#).

The challenge now is to continue to cycle to work. The CERN community is encouraged to cycle all year around, but remember: safety first! Don't forget to check out the [safety rules for cycling](#) and the [road safety pages](#) and to complete the online course [Road Traffic – Bike Riding](#) before getting in the saddle.

## Contents

### News.....p.1-13

Bike to Work 2024: CERN achieves the best participation rate

Accelerator Report: Navigating hot summer days and thunderstorms

CERN70: Green light for LEP

Updating the European Strategy for Particle Physics

CERN Science Gateway featured in TIME magazine's "World's Greatest Places"

News from ICHEP: ATLAS probes uncharted territory with LHC Run 3 data

LHCb investigates the properties of one of physics' most puzzling particles

HiLumi News: new large helium tanks

Digital archaeology: new LEP data now available to all

ATLAS announces its 2024 Outstanding Achievement Award winners

CMS congratulates its 2023 Award and Thesis Award winners

CERN and Pro Helvetia announce the artists selected for the Connect India residency

Computer Security: Security NIMBYs

### Official communications.....p.14-18

2023 Annual Report – Chapter VI of the Staff Rules and Regulations

CERN publishes its first F-Gas Policy

### Announcements.....p.18-22

Upcoming events

REMINDER: Invitation to the CERN70 community event

REMINDER: CERN70 Community Event - Call for volunteers

Winner of the CERN70 T-shirt design competition

The new CERN Community Support Centre

A Particular Score: Listen to cosmic rays at the new CERN Community Support Centre

The Locks and Keys service is moving to Building 73

### Ombud's Corner.....p.22

When disagreements or conflicts arise - don't let them get out of control

# Accelerator Report: Navigating hot summer days and thunderstorms

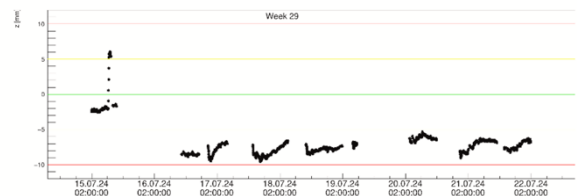
The summer is often a very productive period for physics in CERN's accelerator complex, that is, when the weather remains reasonable...

Although summer weather has finally arrived and many have left or will soon leave for holidays, the accelerator complex has no intention of letting up. In fact, the summer is often a very productive period for physics: in July, the LHC was producing collisions around 55% of the time.

During the summer, the activities scheduled are mainly aimed at physics production, leading to an increased number of cycles for the experiments and fewer interruptions. However, the warm weather, especially when coupled with high humidity, can pose challenges for the cooling of the accelerators. Many of the accelerator systems, including the cryogenics that keep the LHC magnets cold and superconducting, rely on cooling towers, but high outside temperatures and humidity reduce their cooling capacity.

High temperatures can also affect the buildings that house critical equipment, such as power converters, beam instrumentation electronics and radiofrequency electronics. For instance, a recent temperature increase of 2 to 3°C in the LHC surface building 2475, located above LHC Point 4, caused a noticeable shift in the collision point along the longitudinal axis of the beam in all four LHC detectors (see graphic for the ALICE experiment). This shift meant that collisions no longer occurred in the centre of the experiments. Experts from the Cooling and Ventilation group (EN department) intervened, adjusting the temperature regulation to improve the situation. LHC operations experts established that for each degree celsius increase in the building, the collision point shifted by approximately 1.5 mm. This demonstrates the importance of stable

temperatures in these buildings, especially during the hotter days of the summer.



*Position of the collision point in the ALICE detector along the longitudinal beam axis. Between 15 and 16 July, a clear shift of the collision point can be observed, moving from 2.5 mm to about 7 mm away from the centre of the detector. (Image: CERN)*

Another summer challenge is the increased occurrence of thunderstorms. The lightning associated with these storms can cause electrical perturbations, potentially stopping some of the accelerator equipment and thus halting the beam. Interestingly, it is not usually the thunderstorms directly above CERN that cause these issues, but those affecting the areas around the high voltage electrical lines.

Over the years, the equipment groups have worked hard to make the accelerator complex more resilient to electrical perturbations, and their efforts have paid off. However, it is impossible to make all equipment completely insensitive to electrical glitches. So, when you see lights flickering in your house during a thunderstorm, operators and equipment experts at CERN are probably very busy re-establishing beam production in our accelerators.

*Rende Steerenberg*

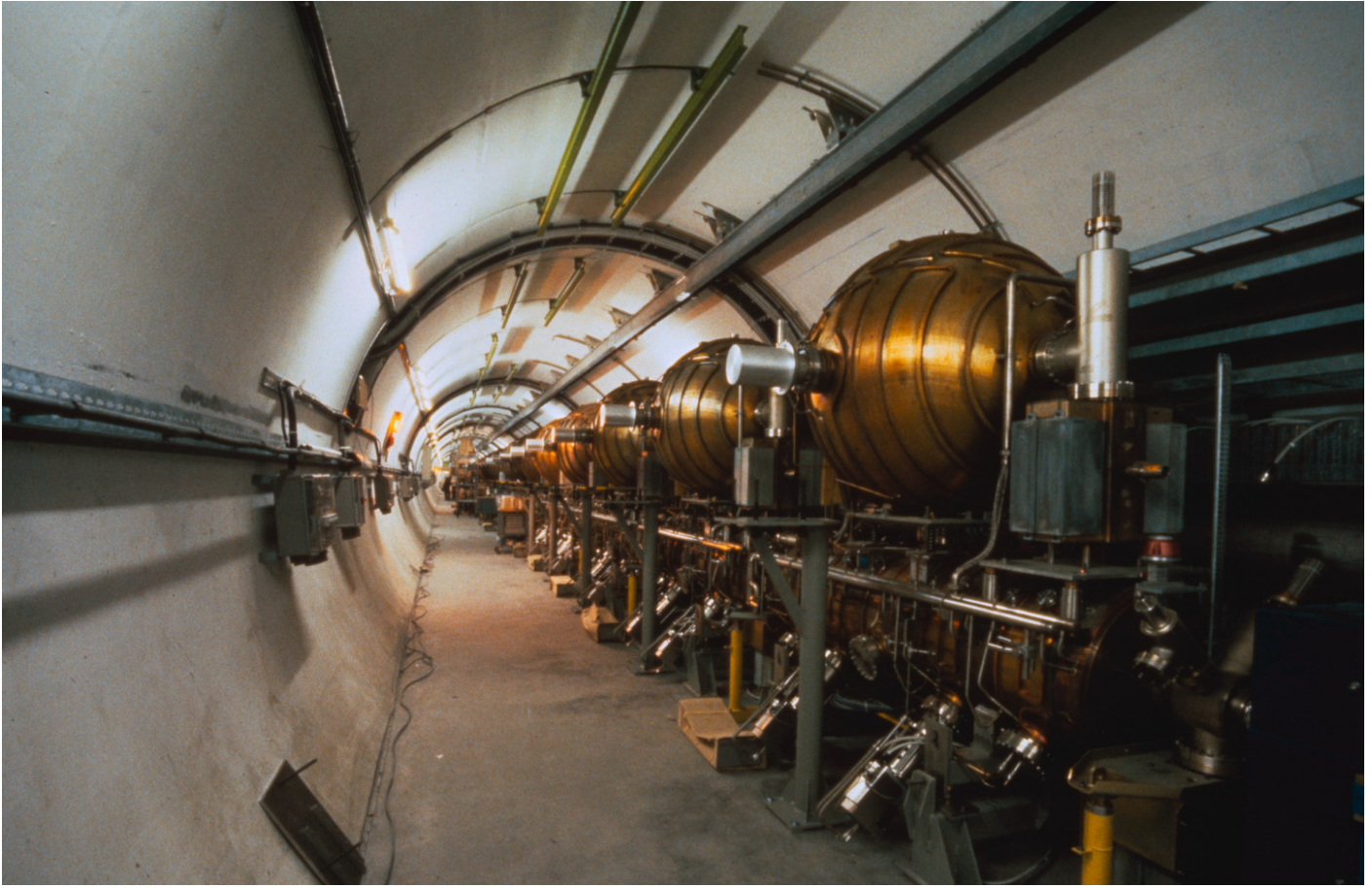
---

## CERN70: Green light for LEP

Part 14 of the [CERN70](#) feature series. Herwig Schopper was Director-General of CERN from 1981 to 1988, during which time the Large Electron Positron collider was approved and constructed

The [Large Electron Positron collider \(LEP\)](#) project was first presented to the CERN Council in June 1980, following the development of several concepts at different energies and lengths. The project envisaged an

energy of 50 gigaelectronvolts (GeV) per beam and a circumference of 27 km, using the [Proton Synchrotron \(PS\)](#) and [Super Proton Synchrotron \(SPS\)](#) as pre-accelerators.



*The Large Electron Positron (LEP) collider operated from 1989 to 2000. It initially used spherical copper radio-frequency cavities, shown here, which were replaced by superconducting accelerating cavities in 1995. (Image: [CERN](#))*

After negotiations with Member States, a final “stripped down version” was adopted in 1981. The idea? To design LEP as an “evolving machine”: after an initial phase as a Z factory ([the Z particle would not be discovered until 2 years later](#)), LEP's energy could, in a second phase, be doubled by incorporating superconducting accelerator cavities.

On 13 September 1983, French President François Mitterrand and his Swiss counterpart Pierre Aubert launched the construction of LEP. Excavated 100 metres below the surface, with a circumference of 27 km, this accelerator was to become the largest in the world. It was the biggest civil engineering project in CERN's history, and the largest European construction site before the Channel Tunnel.

Obviously, choosing the precise location for a subterranean ring between the Jura Mountains and Lake Geneva was a far from simple task, and the route eventually had to pass along the edge of the Jura, a section where the construction work encountered many challenges. For geological reasons, the tunnel was excavated with an inclination of 1.4%, making LEP the first inclined collider. Finally, after a few twists and turns, the project was completed in February 1988.

LEP's other civil engineering feat was its precision. The components of the accelerator, the magnets and cavities, had to be positioned to within 0.1 millimetre of one another. This required excavating a near-perfect tunnel using tools and techniques that were at the cutting-edge of technology.

At the same time as LEP was approved and built, an experimental programme was set up. Interest was considerable, but only four experiments could be approved. The organisation of these experiments, known as the “LEP Model”, was a first in terms of large-scale international collaboration, with groups from all over the world taking part.



## Recollections

Herwig Schopper came to CERN in the early 1970s to head the Nuclear Physics Division. Then, after chairing the directorate of the German nuclear research centre, DESY, for several years, he returned to CERN in 1981. He was Director-General of CERN from 1981 to 1988, during which time the Large Electron Positron (LEP) collider was approved and constructed.



*Herwig Schopper in front of the LEP final plan in 1982. (Image: CERN)*

“My first personal experience with LEP was a rigorous examination in the Committee of Council. One delegation was against my nomination as Director-General, suspecting that I would favour the German DESY site for LEP instead of CERN. After I had explained my intentions, the delegate concerned received new instructions by telephone during a coffee break; I was elected unanimously and the approval procedure for LEP could start. While the approval of LEP was still pending, Margaret Thatcher visited CERN. On her arrival she told me that she wanted to be treated as a fellow scientist and not as Prime Minister. She surprised me with the question of why we intended to build a circular machine instead of two opposing linear colliders, a very pertinent question and proving her excellent briefing for the visit. Indeed, it is a delicate matter of optimisation which of the two options is more favourable for a certain beam energy. I explained to her that in the case of LEP a circular machine was more cost

effective. She accepted the argument and asked how big the tunnel would be for the next project after LEP. To my reply that the LEP tunnel would be the last ring at CERN she retorted: “Why should I believe you? When I visited CERN the first time John Adams told me that the SPS tunnel would be the last.” Nevertheless, she stated at a press conference that she had been convinced that the funds at CERN were used efficiently, and subsequently the United Kingdom approved LEP. In the beginning, LEP met considerable opposition in the nearby population. We discovered that this was due to a lack of information, many people believing that the ‘N’ in CERN had something to do with nuclear power. To improve this situation, many senior CERN staff members spent an incredible amount of time giving lectures in all the neighbouring villages.



*One of the four LEP monorail trains suspended from the ceiling of the 27-km long LEP tunnel. Such trains circulate in the tunnel transporting materials and workers. The Swiss and French customs officers were photographed in the tunnel to symbolise that the LEP tunnel crossed the French-Swiss border in six different places. (Image: Peter Ginter/CERN)*

However, I was particularly worried when Denis de Rougemont, a well-known Swiss writer, published articles claiming that LEP was inhuman, gigantic and not necessary for society. This was a surprise since de Rougemont had founded the “Centre européen de la culture” immediately after the war, where politicians discussed how European scientists could learn to work peacefully together again. This political initiative, little known, was one of the channels that led to the foundation of CERN, the other one originating from physicists. I invited de Rougemont to lunch where we discussed his objections; in the end we resolved the difficulties,



and I promised to explain his essential role for the creation of CERN at a public debate. We always tried to take decisions on the basis of rational arguments. However, in the case of deciding on the location of the two experiments ALEPH and DELPHI, this method failed. So, it was agreed that a coin would be flipped. At a Directorate meeting, this act was performed in the presence of the two spokespeople, Jack Steinberger and Ugo Amaldi. After tossing the coin, Jack, being a very careful experimentalist,

checked whether the coin had indeed two different sides.”

----

*This interview is adapted from the 2004 book “Infinitely CERN”, published to celebrate CERN’s 50th anniversary. Schopper recently [celebrated his 100th birthday](#), watch the [recording of the dedicated symposium at CERN](#) and read his recently published [open access book](#) that is both a memoir and a biography.*

## Updating the European Strategy for Particle Physics

Important information and key dates for the 2024-26 update of the European Strategy for Particle Physics



On 21 March, the CERN Council formally launched the update of the European Strategy for Particle Physics, a two-year process involving the whole community that aims to develop a common vision for the future of particle physics in Europe within the international context. The process is expected to be concluded in June 2026 with the approval of the updated Strategy by the Council.

According to the remit of the European Strategy Group (ESG), the aim of the Strategy update is to develop a visionary and concrete plan that greatly advances human knowledge in fundamental physics through the realisation of the next flagship project at CERN; this plan should attract and recognise the value of international collaboration and should allow Europe to continue to play a leading role in the field.

In addition to the input of the particle physics community, the ESG will take into consideration: the status of implementation of the 2020 Strategy update, accomplishments over recent years (including results from the [LHC](#) and other experiments and facilities worldwide, progress in the construction of the [High-Luminosity LHC](#), the

outcome of the [Future Circular Collider](#) Feasibility Study and recent technological developments in accelerator, detector and computing technologies) and the international landscape.

The Strategy update should include the preferred option for the next collider at CERN and prioritised alternatives to be pursued if the chosen preferred option turns out not to be feasible or competitive. It should also indicate priorities for areas of exploration that are complementary to colliders, for other experiments to be considered at CERN and at other laboratories in Europe, as well as for participation in projects outside Europe.

Finally, the ESG will review and update the Strategy and add other items identified as relevant to the field, including: the theory frontier; actions to minimise the environmental impact and improve the sustainability of accelerator-based particle physics; initiatives to attract, train and retain the young generations; and public engagement.

To inform this vital process, the Secretariat of the ESG is calling on particle physicists across universities, laboratories and national institutes to provide input to the process in various forms and at different stages. Details of the organisation and timeline of the full process and how to submit input can be found at: <https://europeanstrategyupdate.web.cern.ch>

In a first step, the particle physics community is invited to submit written input by **31 March 2025**. The portal for the submission, and for follow-up

submissions later on, will be via the web page. More details will be communicated later.

This first step will be followed by a Scientific **Open Symposium** from 23 to 27 June 2025, at which the community will be invited to debate the future orientation of European particle physics. A “**Briefing Book**” based on the input and discussions will then be prepared by the Physics Preparatory Group (PPG), which will be established by the Council in September 2024. The Briefing Book will be submitted to the ESG by the end of September 2025 for consideration during a

five-day-long Drafting Session, which is scheduled to take place from 1 to 5 December 2025.

To allow the **national communities** to react to the submissions collected by March 2025 and to the content of the Briefing Book, the ESG will offer further opportunities for input ahead of the Open Symposium (with a deadline of 26 May 2025) and ahead of the Drafting Session (with a deadline of 14 November 2025).

“We look forward to a very fruitful process over the forthcoming one and a half years,” states Karl Jakobs, University of Freiburg, Chair of the Strategy Secretariat.

---

## CERN Science Gateway featured in TIME magazine’s “World’s Greatest Places”



(Image: CERN)

CERN Science Gateway is one of the [100 greatest places](#) to visit in the world in 2024, according to

TIME magazine. Published last week, the magazine’s special edition [praises](#) the Laboratory’s centre for education and outreach for its variety of exhibitions and activities on offer for all ages, as well as the facility’s unique artworks and architecture.

This kind of achievement is made possible thanks to the help of all the CERN Science Gateway guides, who make it an authentic and welcoming space for visitors. Find out more about becoming a CERN guide [here](#).

CERN Science Gateway is open from Tuesday to Sunday, from 9.00 a.m. to 5.00 p.m. (last entry at 4.30 p.m.). Find out about all that Science Gateway has to offer on this [website](#).

---

## News from ICHEP: ATLAS probes uncharted territory with LHC Run 3 data

The ATLAS collaboration has released its first results from searches for new physics phenomena conducted with data from Run 3 of the LHC

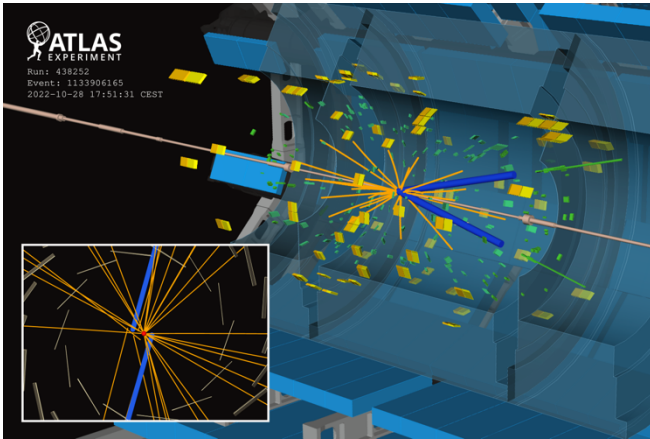
Despite its immense success in describing the fundamental building blocks of matter and their interactions, the [Standard Model](#) of particle physics is known to be incomplete. Experiments around the globe and in space are therefore

searching for signs of new physics phenomena that would guide physicists towards a more comprehensive theory.

At the biannual [ICHEP](#) conference that took place in Prague earlier this week, the [ATLAS](#)



collaboration presented its first results from searches for new physics at record collision energies, targeting magnetic monopoles produced in heavy-ion collisions and long-lived particles created in proton–proton collisions.



*Display of a collision event recorded by the ATLAS detector at an energy of 13.6 TeV, featuring two candidate displaced electrons each represented by a track (blue lines) that points to an energy deposit in the ATLAS calorimeter (green). The inset shows an axial view of the detector, illustrating the electron tracks displaced from the interaction point (red circle) by a few mm. (Image: ATLAS/CERN)*

Magnetic monopoles are hypothetical particles with only a single north or south pole, making them magnetically charged. Their existence would demonstrate the complete symmetry between electricity and magnetism. It would also confirm aspects of “grand unified theories” beyond the Standard Model that unify the strong, weak and electromagnetic forces at very high energies.

Researchers at the [Large Hadron Collider](#) (LHC) are searching for monopoles produced in high-energy collisions. Monopoles would be highly ionising, meaning they would strip electrons from atoms and would leave behind significant energy deposits in particle detectors.

In a [new search for magnetic monopoles](#), the ATLAS collaboration analysed its first heavy-ion (lead–lead) collision data from LHC Run 3, which was collected in the autumn of 2023 at an unprecedentedly high energy of 5.36 TeV per pair of nucleons (protons or neutrons). Specifically, ATLAS researchers looked at ultraperipheral collisions, in which the ions do not collide centrally via the short-ranged strong interaction, but instead pass by close enough to interact through the weaker but long-ranged electromagnetic force. Collisions among lead ions can produce the

largest magnetic fields in the Universe, with strength up to 1016 Tesla.

If a pair of magnetic monopoles was [produced in such interactions](#), it would be the sole particle system to be found in an otherwise empty detector, and it would manifest itself as a concentrated cloud of ionisation electrons. Looking for the unique signal features and analysing backgrounds that could mimic them, ATLAS saw no signs of monopoles in their Run 3 heavy-ion data.

Consequently, the result sets the world's best limits on the production rate of monopoles created in ultraperipheral heavy-ion collisions for monopole masses below 120 GeV. Moreover, this analysis introduces a methodology for studying highly ionising particles in heavy-ion data from the LHC and beyond.

Most searches for new physics look for new particles that would decay “promptly” and produce decay products that emanate from the LHC’s proton–proton interaction points. However, beyond-the-Standard-Model physics theories, including supersymmetry, also predict “long-lived particles” that would produce decay products away from the interaction point. Such particles require dedicated techniques to reconstruct particle tracks and may have eluded detection in prior searches.

ATLAS has [released](#) the result of a new search for a pair of long-lived particles, each of which decays into an electron, muon or tau lepton, resulting in two particle tracks that are “displaced” from the ATLAS interaction point (see image above) – a rare signature that could be indicative of new physics. In particular, ATLAS looked for a new signature where one of the long-lived particles travels far enough before decaying so that only a single electron is detected.

This is the first ATLAS search of this type using the 13.6 TeV proton–proton collision data from LHC’s Run 3. In preparation of Run 3, ATLAS researchers had enhanced the online collision-event selection – the “trigger” – with the reconstruction of displaced tracks, which enabled the present search for new long-lived particles.

The event yields in all search regions matched Standard-Model expectations. These results set the strictest limits yet on the long-lived

supersymmetric partners of electrons, muons, and tau leptons.

With more data from the LHC and its future upgrade, the [High-Luminosity LHC](#), ATLAS physicists will continue their quest to find long-lived particles, magnetic monopoles and other

hypothetical particles – all while further refining their search techniques and developing new experimental strategies.

Find out more [here](#) and [here](#).

ATLAS collaboration

---

## LHCb investigates the properties of one of physics' most puzzling particles

**The particle, known as  $\chi_{c1}(3872)$ , has fascinated physicists for years. Now, the LHCb collaboration is closer to finding out what it is made up of**

$\chi_{c1}(3872)$  is an intriguing particle. It was first discovered over 20 years ago in  $B^+$  meson decays by the [BELLE collaboration](#), KEK, Japan. Since then, the [LHCb collaboration](#) reported it in 2010 and has measured some of its properties. But here's the catch – physicists still don't know what it is actually made up of.

In the [quark model](#) of particle physics, there are baryons (made up of three quarks), mesons (made up of a quark–antiquark pair) and exotic particles (made up of an unconventional number of quarks). To find out what  $\chi_{c1}(3872)$  consists of, physicists must measure its properties, such as its mass or quantum number. Theories suggest that  $\chi_{c1}(3872)$  could be a conventional charmonium state, made up of charm and anticharm quarks, or an exotic particle composed of four quarks. An exotic particle of this type could be a tightly bound tetraquark, a molecular state, a  $cc$ -gluon hybrid state, a vector glueball or a mixture of different possibilities.

Previously, the LHCb collaboration has found its [quantum number](#) to be  $1^{++}$  and, in 2020, made [precise measurements](#) of the width (lifetime) and mass of the particle. The collaboration also measured what is known as its low-energy scattering parameters. The results showed that its mass is just a tad smaller than the sum of the masses of the  $D^0$  and  $D^{*0}$  mesons.

Following these results, the theoretical community was divided. Some argued that  $\chi_{c1}(3872)$  was a molecular state consisting of spatially separated  $D^0$  and  $D^{*0}$  mesons. This molecular state would be much larger than the typical size of particles and more comparable to a heavy nucleus. However, this argument

encounters a problem, namely that physicists expect molecular objects to be suppressed in hadron–hadron collisions, and the  $\chi_{c1}(3872)$  is produced abundantly. Other theorists interpreted the results as clear evidence that  $\chi_{c1}(3872)$  has a “compact” component. This would mean it is a particle with much smaller size, containing either a tightly bound charmonium or a tetraquark.

One way to help determine what  $\chi_{c1}(3872)$  contains is to calculate the ratio between probabilities of the decays into different lighter particles (branching fractions). By comparing the rate at which it decays either to an excited charmonium state or to a charmonium state and a photon, physicists can gather clues as to what type of particle it is. There is a clear theoretical signature: if the ratio is non-vanishing, it is evidence for some compact component in  $\chi_{c1}(3872)$ , disfavours the pure molecular model. Now, using the complete set of LHC Run 1 and Run 2 data, the LHCb collaboration has found these ratios to be non-vanishing, with a significance exceeding six standard deviations. The large measured value of the ratios is inconsistent with the expectations based on the pure  $D^0D^{*0}$  molecular hypothesis for the  $\chi_{c1}(3872)$  particle. Instead, it supports a wide range of predictions based on other hypotheses of the  $\chi_{c1}(3872)$  structure, including conventional (compact) charmonium, a compact tetraquark containing a charm quark, charm antiquark, light quark and light antiquark, or a mixture of molecules with a substantial compact core component. In short, the result provides a strong argument in favour of the  $\chi_{c1}(3872)$  structure containing a compact component.

---

## HiLumi News: new large helium tanks

Four large helium tanks will supply the new High-Luminosity LHC refrigerators



Two large helium tanks for the High-Luminosity LHC were installed at Point 1 in June. (Image: CERN)

Exceptional machines call for exceptional operations. Two large helium tanks for the High-Luminosity LHC (HL-LHC) were installed at Point 1 in June, and two more at Point 5 in July. They will store the helium for the refrigerators that will cool

the [HL-LHC's new focusing magnets](#) on both sides of the ATLAS and CMS experiments.

Each tank weighs over 62 tonnes and is 28 metres long and 3.5 metres in diameter. The tanks will each be able to store 250 cubic metres of gaseous helium at a pressure of 20 bars at room temperature, representing a weight of around 800 kg. As the tanks were manufactured in Portugal, it took more than eight days to transport them to CERN under escort as an abnormal load.

The new refrigerators – one for each point – will be delivered next year. “We have a good year of work ahead of us to install the complete infrastructure and connect the tanks to the helium distribution system,” explains Antonio Suraci from the Cryogenics group.

Around 130 tonnes of helium are needed to cool the superconducting magnets of the LHC and its experiments. When the HL-LHC is up and running, it will consume almost the same amount of helium, but the design of the cryogenic system will have to be modified to supply this new equipment on each side of the ATLAS and CMS experiments.

---

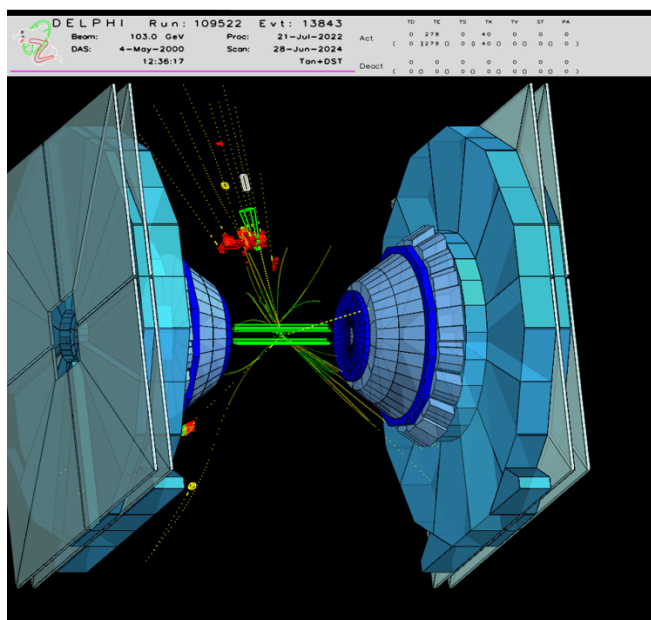
## Digital archaeology: new LEP data now available to all

Retrieving and preserving access to data from experiments that ran in the 90s is a complex task carried out by passionate experts in the IT department

Unlike letters carved on the Rosetta stone, digital data is not written on a virtually immutable support. Just a few years after it is written, its format becomes obsolete, the readout analysis tools can't run on computers and the visualisation code no longer works. But data can still contain interesting scientific information that should remain available to future generations of scientists.

A set of data of potentially high interest is that of LEP, CERN's former flagship accelerator that collided electrons and positrons up until 2000. Like the current LHC, LEP had four collision points, each hosting an experiment – ALEPH, DELPHI, OPAL and L3 – that was operated by hundreds of scientists. LEP holds the record for the world's highest e+e- energy collisions but the data collected over two decades ago remains available to only a small community of people.





Collision data from the DELPHI experiment at the Large Electron Positron Collider. (Image: CERN)

Like archaeologists who unearth the remnants of past civilizations, digital archaeologists are computing experts who retrieve data years after the collaborations have moved on to other experiments. “The first step is to reach agreement within the collaboration as to opening and sharing their data and the software required to exploit it. Then, just like archaeologists, we dig into the documents that the former collaborations have written about the data architecture, and retrieve the software used for actual analysis”, explains Ulrich Schwickerath, a former DELPHI physicist and computing expert working in the IT department. This is no easy task because the

information often lies in unpublished documents or in private repositories that might have not even been shared within the collaboration.

The analysis software from LEP times was deposited in CERNLIB, a CERN-developed software library that was discontinued in 2003. “Shortly after the last release of CERNLIB, many external enthusiasts kept it alive and applied quick fixes to the software, known as patches. In a community-based effort, these patches were gathered together in order to create a community version, allowing the old software to be adapted to modern architectures,” Ulrich explains. “Since then, together with a few LEP enthusiasts, we have managed to resurrect the software stacks of the DELPHI and OPAL experiments using the new community-driven version of CERNLIB. We are working towards making the dataset fully available in the original format, as compatible as possible with modern hardware and software tools, and to revise the old visualisation codes so that today’s scientists can run proper analyses.”

The data from ALEPH and DELPHI is now open, and the DELPHI data will soon become available on the [CERN Open Data Portal](#). So whether you’re a researcher, teacher, student or just an interested non-physicist, start your discovery of electron-positron annihilation data with the DELPHI detector by visiting [this webpage](#).

Antonella Del Rosso

## ATLAS announces its 2024 Outstanding Achievement Award winners



Winners of the 2024 ATLAS Outstanding Achievement Awards. (Image: ATLAS collaboration)

The ATLAS collaboration held its seventh biennial Outstanding Achievement Awards ceremony on 20 June 2024. These awards recognise the invaluable technical work performed across the collaboration in various fields. This year, the Awards Committee honoured two individuals and seven groups for their exceptional contributions to detector operation, detector upgrades, software, computing and teamwork during the period from February 2022 to October 2023.

“The work of the award winners demanded great creativity and determination,” said Sarah Demers,

co-chair of the Awards Committee. “Not only were their contributions crucial during this recent data-taking period, but they also set the stage for high-quality physics analyses and operations in the years to come.”

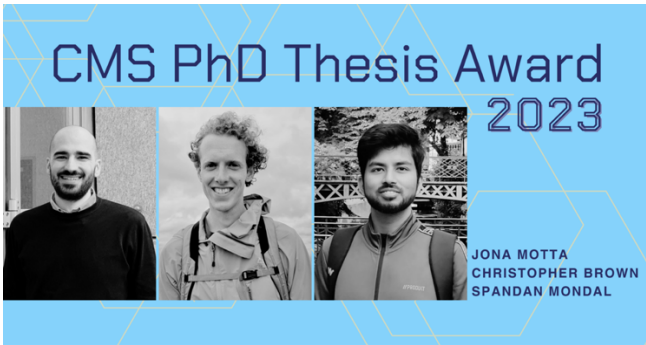
Find out about the award winners on the [ATLAS website](#).

*ATLAS collaboration*

---

## CMS congratulates its 2023 Award and Thesis Award winners

CMS PhD Thesis Award winners 2023



Each year, the CMS collaboration honours the work of exceptional PhD students with the Thesis Award. This award recognises doctoral research conducted within the collaboration that pushes the boundaries of high-energy physics. Out of a highly competitive pool of 27 nominees, the winners of the 2023 CMS PhD Thesis Award are Jona Motta (LLR, Institut Polytechnique de Paris), Christopher Edward Brown (Imperial College London) and Spandan Mondal (RWTH Aachen University, Germany).

“Doctoral students do a lot of impressive work in CMS. Writing a PhD thesis to document this work is a tremendous effort and achievement. Some of the students decide to invest substantial extra efforts in writing an exceptionally clear, effective and original documentation of their research work. They write for their peers, future students, who will follow in their steps, and for all those in search of in-depth, detailed, accurate but also accessible knowledge related to CMS scientific or technical frontline research,” says Marta Felcini, chair of the CMS PhD Thesis Award Committee. Read more about the winners on the [CMS website](#).

CMS Award winners 2023



Each year since 2000, the CMS Awards Committee has recognised outstanding contributions from members of the CMS collaboration, honouring their dedication to the experiment. Nominations can be made by any CMS member for exceptional work in various fields, and the winners are selected by a dedicated committee.

“This year, the CMS Awards celebrate the hard work that was done by our collaborators both on the operations and on the upgrades to ensure the success of the present and future detector,” say the CMS Awards Committee Chairs.

As well as the 50 award winners, the RD53 collaboration was singled out by CMS for special recognition. This dedicated group is responsible for the development of the ATLAS and CMS Phase 2 [inner tracker readout](#) chips.

Find out more about the collaboration and each of the 50 awardees on the [CMS website](#).

*CMS collaboration*

# CERN and Pro Helvetia announce the artists selected for the Connect India residency

Arts at CERN, in collaboration with the Swiss Arts Council Pro Helvetia, announced today that Swiss artist Lou Masduraud and Indian artist Shailesh BR have been selected for the Connect India dual residency

Dual residency at  
CERN and ICTS

Connect  
India



Indian artist Shailesh BR and Swiss artist Lou Masduraud are the two selected artists for the Connect India dual residency. (Image: CERN)

[Connect](#) is an art residency programme launched by [Arts at CERN](#) and [Pro Helvetia](#) in 2021, which serves as a platform to foster experimentation in art and science by bringing artists into contact with fundamental science and cutting-edge research at CERN and other international scientific organisations.

The two awardees, [Lou Masduraud](#) and [Shailesh BR](#), selected by a board of cultural experts, will be invited to a three-week residency at the [International Centre for Theoretical Sciences](#) (ICTS) in Bengaluru, followed by a three-week stay at [CERN](#) in Geneva. They will receive support from the Arts at CERN and ICTS curatorial teams to explore new forms of artistic expression and transform these explorations into art productions.

Shailesh BR is a visual artist based in Delhi NCR, India. His practice explores fundamental aspects

of our world by examining existing knowledge, systems, traditions and philosophical thoughts. Utilising a diverse visual vocabulary, he incorporates methods from science and technology into his artistic interventions, creating drawings, object modifications and machines.

Lou Masduraud lives and works in Geneva. Through sculpture and installation, Masduraud proposes alternative narratives to dominant realities. Her body of work explores the intricate network of human activities through formal and material investigations of everyday elements such as fountains, basement windows and pipes, in both public and private spaces.

Connect India offers these artists a unique opportunity to engage with cutting-edge fundamental scientific research in both Geneva and Bengaluru. At CERN, physicists and engineers design and use a wide array of experiments in particle physics, while ICTS focuses on theoretical sciences, encompassing fields like physics, mathematics, astronomy and computational biology.

Connect India marks the second collaboration with ICTS, following successful dual residencies at scientific organisations in Chile and South Africa. Since its launch in 2021, the Connect collaboration between Pro Helvetia and Arts at CERN has hosted seven editions and established itself as a key platform for exchange between communities of artists and scientists around the world.

---

## Computer Security: Security NIMBYs

Security sucks. No doubt about it. It's a cost factor, a resource drain, and does not provide any immediate benefit. It creates hurdles, inconveniences and complications. The same goes for driving licences, speed limits, insurance and

any other kind of regulation. Security sucks like the nuclear power plant in view, the wind turbines or motorways within hearing distance or any kind of vaccination. However, while we all agree (don't we?) that the existence of driving licences, speed



limits, motorways, insurance, nuclear power plants, wind turbines, vaccinations and even “security” is justified, we feel that, ideally, it should not directly affect or impact us. NIMBY – “not in my backyard”.

From the zillions of discussions we’ve had since the deployment of the new [outer perimeter firewall](#), the new [mail quarantining appliance](#) and, with it, the roll-out of new antispooofing measures and the [recent activation of two-factor authentication](#), it’s clear that NIMBYs also exist in our community. Blocking malicious top-level domains? Yes, but don’t block the webpage of my personal yoga trainer. Protecting against impersonation? Sure, but don’t touch my private emails sent to my professional @cern.ch email address. Two-factor authentication? Great idea, but please don’t start with my department. And we have many more examples, with some even suggesting that we are intentionally inhibiting CERN’s functioning, hate certain users (communities) and do not deserve to get any promotions this year.

If, however, we want to protect CERN and to have a decent level of security, we cannot avoid deploying a certain set of computer security measures: defence-in-depth, network segregation, firewalls, intrusion prevention, SPAM and malware filtering, endpoint protection, strong passwords and two-factor authentication, secure software development mechanisms, business continuity and disaster recovery planning. And they all come with inconveniences. They reduce flexibility and spontaneity. They create choke points (e.g. firewalls, mail filters, software scanners) with security controls to distinguish

good from bad. Like fuses, water filtration, traffic lights, cashiers or border control. And it would be hilarious and/or unfair to argue that “[some animals are more equal than others](#)” and to provide them with exemptions, derogations or any other kind of exceptions. No NIMBYs.

The same goes for CERN computer security. Security paradigms and deployment mean following a simple priority list: high-risk areas first, critical functions first, internet-facing services first, professional business first. Everything else is secondary and must follow the rules for the “firsts”. Ideally with no exceptions as they might lead to security bypasses or introduce weaknesses or vulnerabilities. A dedicated cybersecurity audit confirmed our strategy and requested that we enforce it more thoroughly (more about this in a future Bulletin article). So, bear with us, if your login takes a little longer, while some of your colleagues can still “enjoy” one-factor authentication. Bear with us, if your personal emails got quarantined as they are violating industrial standards and are released only later. Bear with us, if your software compiles with (too) many error messages and requires fixing before you can use it. Sorry, NIMBYs.

Security sucks. Indeed. But the alternative sucks even more. [Reinstalling your laptop](#). [Exposing all your emails](#). [Losing all your documents](#). [Sabotaging your system, service or experiment](#). [Ransomware the Organization](#). [Stopping CERN from working](#). [For a looooong time](#). Hence: RIP NIMBYs.

*The Computer Security team*

# Official News

*Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.*

## 2023 Annual Report – Chapter VI of the Staff Rules and Regulations

### Settlement of disputes and discipline

The Organization is committed to a fair and respectful work environment. Conduct issues or administrative disputes brought to the attention of the Organization are addressed in a timely manner using, whenever possible, informal resolution mechanisms such as mediation. In cases where informal resolution is not achievable or appropriate, the Organization or the member of the personnel concerned may decide to initiate formal proceedings under the Organization's settlement of disputes procedure or conduct-related frameworks, as applicable [1]. This report provides an overview of the cases handled under Chapter VI of the Staff Rules and Regulations in 2023.

#### Introduction

The Annual Report under Chapter VI ("Settlement of Disputes and Discipline") of the Staff Rules and Regulations serves to report on:

- requests for review;
- internal appeals;
- cases in which disciplinary action was taken; and
- complaints before the Administrative Tribunal of the International Labour Organization (ILOAT).

#### Requests for review and internal appeals

Under Article S VI 1.01 of the Staff Rules, members of the personnel may challenge an administrative decision by the Director-General where it adversely affects the conditions of employment or association that derive from their contract or from the Staff Rules and Regulations.

If permitted by the Staff Rules and Regulations, a decision may be challenged internally within the Organization:

- through a review procedure; or
- through an internal appeal procedure. In this case, the Joint Advisory Appeals Board (JAAB) [2] shall be consulted by the Director-General prior to taking any final decision on the merits of the appeal.

#### Disciplinary Action

Under Article S VI 2.01 of the Staff Rules, the Director-General may take disciplinary action against members of the personnel who, whether intentionally or through carelessness, are guilty of a breach of the Rules and Regulations or of misconduct that is to the detriment of the Organization.

Article S VI 2.02 of the Staff Rules stipulates that, having regard to the gravity of the breach or misconduct in question, the disciplinary action shall be:

- a warning;
- a reprimand;
- suspension without remuneration or pay for a period not exceeding six months;
- downward adjustment of the staff member's salary;
- demotion;
- dismissal.

The Director-General shall consult the Joint Advisory Disciplinary Board (JADB [3]) prior to taking any disciplinary action other than a warning or a reprimand (Article S VI 2.04 of the Staff Rules). In cases of particularly serious misconduct, the Director-General may decide to dismiss

without notice and without consulting the JADB (Article S VI 2.05 of the Staff Rules).

### **Complaints before the Administrative Tribunal of the International Labour Organization (ILOAT)**

A decision may be challenged externally through the filing of a complaint before the ILOAT:

- when internal procedures have been exhausted and the decision is final;
- when an internal challenge is not permitted by the Staff Rules and Regulations; or
- when the complainant is authorised by the Director-General to proceed directly to the Tribunal.

\*\*\*\*

### **A - Requests for review:**

From 1 January to 31 December 2023, five requests for a review of administrative decisions were introduced:

- i. In May 2023, a fellow [\[4\]](#) requested the review of the decision not to pay compensation for untaken annual leave on termination of their employment contract at the end of the probation period. In addition, the fellow requested that a decision be taken to award financial compensation for loss of earnings beyond the end of their contract and to make efforts concerning rehabilitation. The Director for Finance and Human Resources decided to maintain the decision concerning the refusal of compensation for untaken leave and rejected the two additional requests.
- ii. In June 2023, a staff member requested the review of the decision not to be granted a lump-sum payment for family-related travel expenses in connection with the loss of a close relative, because the request had been submitted outside the regulatory time limit. Given the exceptional circumstances of the request however, and that the staff member had not been adequately informed about the procedure, an exceptional exemption from the time limit was granted allowing the payment to be made.
- iii. In September 2023, a staff member requested the review of the decision to classify an accident they suffered during a private trip following a duty trip as being of a non-occupational nature. The Director for Finance and Human Resources decided to uphold the decision: an accident occurring during a private trip

following a duty trip cannot be of an occupational nature.

iv. In November 2023, a former fellow requested the review of the decision not to grant travel expenses for their spouse when they took up their appointment at CERN. The Director for Finance and Human Resources decided to uphold the decision as the regulatory criteria for payment were not met.

v. In December 2023, a former fellow requested the review of the decision taken by the Director-General, further to an investigation by the Harassment Investigation Subpanel, to reject a complaint for harassment on the basis that no element of harassing behaviour had been detected. The request for review was withdrawn following amicable resolution.

### **B - Internal appeals (Joint Advisory Appeals Board (JAAB)):**

During the period from 1 January to 31 December 2023, seven internal appeals were introduced:

- i. In March 2023, three staff members introduced internal appeals against the decision of the CERN Council to apply a 2.5% crisis levy to the staff members' basic salaries in 2023 compensated by five additional days of annual leave to be taken during the year. All three staff members decided to withdraw their appeals.
- ii. In March 2023, a staff member introduced an internal appeal against the decision taken by the French Ministry for Europe and Foreign Affairs to reject the Organization's request to register the staff member's private vehicle on special series plates (green plates). The Director for Finance and Human Resources considered the appeal irreceivable because the contested decision was not a decision taken by the Director-General but taken by the French Ministry. The appellant's request for financial compensation for financial loss was also refused. To be noted that the French Ministry subsequently issued a favourable decision regarding the request for registration.
- iii. In May 2023, a staff member introduced an internal appeal against the decision to transfer them to another unit in their department and qualify their performance as "fair" for the reference year 2022. The appeal was withdrawn following amicable resolution.



iv. In June 2023, a staff member introduced an internal appeal against the decision to qualify their performance as “fair” for the reference year 2022, whilst raising, inter alia, harassment allegations. The appeal was suspended pending the outcome of an investigation by the Harassment Investigation Subpanel. Thereafter, the staff member decided to withdraw the appeal.

v. In August 2023, a former fellow introduced an internal appeal against a review decision not to pay compensation for untaken annual leave on termination of their employment contract at the end of the probation period, not to award financial compensation for loss of earnings beyond the end of their contract, and not to initiate a procedure before the Joint Advisory Rehabilitation and Disability Board with a view to rehabilitation measures [5]. The appeal was withdrawn following amicable resolution.

Concerning previous appeals:

vi. In June 2022, two staff members introduced internal appeals against the Director-General’s decision, which concluded an arbitration procedure, in accordance with the working procedures of the Standing Concertation Committee, concerning the methodology of calculation of the salary increase annual budget in the context of the 2022 MERIT Guidelines; and against the decisions to qualify their individual performance during the year of 2021 as “strong”. Both appellants also requested the Director-General’s authorisation to directly challenge the decisions before the ILOAT, which the Director-General rejected. The Director for Finance and Human Resources considered the appeals receivable only regarding the individual performance qualification decisions and resulting performance rewards. In May 2023, the Director-General decided to follow the recommendation of the JAAB and rejected both appeals.

vii. In July 2021, a staff member introduced an internal appeal against the decision to qualify their performance as “fair” for the reference year 2020, and at the same time raised, inter alia, harassment allegations. This procedure was suspended pending the outcome of an investigation of the Harassment Investigation Subpanel. The internal appeal procedure resumed in January 2023. In November 2023 the Director-General decided to

follow the recommendation of the JAAB and rejected the appeal.

### ***C - Warnings and reprimands:***

In 2023, the Organization issued one warning:

i. A warning was issued to a user who failed to comply with a request by Swiss customs officers to submit to a customs control of their vehicle before entering the site at Gate E. This led to the Swiss Federal Office of Customs and Border Security contacting CERN about the incident.

In 2023, the Organization issued two reprimands:

ii. A reprimand was issued to a staff member further to the outcome of an investigation by the Harassment Investigation Subpanel. The investigation concluded that the staff member had engaged in inappropriate behaviour towards a colleague.

iii. A reprimand was issued to an associated member of personnel who repeatedly falsified their the CERN supervisor’s signature.

### ***D - The Joint Advisory Disciplinary Board (JADB):***

The JADB was not convened in 2023.

### ***E - Dismissal notified during the probation period:***

In 2023, the contracts of two employed members of the personnel were not confirmed following the probation period (as per Article S II 5.01 g) of the Staff Rules).

### ***F - Particularly serious misconduct:***

No contracts were terminated for reasons of particularly serious misconduct in 2023.

### ***G - Complaints before the Administrative Tribunal of the International Labour Organization (ILOAT) [6]:***

During the period from 1 January to 31 December 2023:

#### Complaints introduced:

i. In March 2023, a former staff member filed a complaint against the decision to calculate their indemnity for physical deterioration following an occupational accident at 15%. The Tribunal’s ruling is expected in 2024.

ii. In July 2023, a staff member filed a complaint against the Director-General’s decision to refuse the recognition of total disability resulting in an

incapacity to work. The Tribunal has since suspended the matter on the request of the complainant pending the completion of a new procedure before the Joint Advisory Rehabilitation and Disability Board (JARDB).

iii. In July 2023, a former fellow filed a complaint with the ILOAT against the decision of the Director for Finance and Human Resources to terminate their employment contract at the end of the probation period, due to unsatisfactory performance. The complaint has been withdrawn following amicable resolution.

iv. In August 2023, two staff members each filed a complaint with the ILOAT against the Director-General's decision to dismiss their internal appeals against the individual administrative decisions on the qualification of their performance for 2021 (strong) and the resulting performance rewards, in the context of the 2022 MERIT Guidelines. The Tribunal's ruling is expected in 2026.

#### Judgements issued:

The ILOAT has ruled in four cases involving the Organization, which had been filed in 2021:

v. In two cases filed individually by two staff members requesting the ILOAT to review Judgments 4273 and 4274, the Tribunal dismissed these requests in their entirety by Judgments 4706 and 4705 of 7 July 2023.

vi. In three cases filed individually by three associated members of the personnel against the determination and application of a ceiling on their subsistence allowance payments, which were processed by CERN on behalf of third parties, the Tribunal dismissed the complaints in their entirety by Judgment 4707 of 7 July 2023.

vii. In March 2021, a staff member filed a complaint with the ILOAT against the Director-General's decision to deny their request for reclassification. The Tribunal dismissed the complaint by Judgment 4740 of 31 January 2024.

#### Ongoing complaints:

viii. In April 2020, a former staff member filed a complaint with the ILOAT against the Director-General's decision not to grant them an indefinite contract (IC). The Tribunal's ruling is expected in 2024.

ix. In July and August 2020, four associated members of the personnel filed individual

complaints with the ILOAT against the decision to replace the 2019 internal tax annual certificate by an individual annual statement. The Tribunal's ruling on these four cases is expected in 2025. Since the complainants challenge a decision of general application, should they be successful in their complaints, the outcome would be applied to others in like circumstances [\[7\]](#).

x. In January 2021, a staff member filed a complaint with the ILOAT against the Director-General's decision to reject their internal appeal concerning an allegation of harassment. The Tribunal's ruling is expected in 2024.

xi. In February 2021, a staff member filed a complaint with the ILOAT against the Director-General's decision to maintain their initial performance qualification (fair). The Tribunal's ruling is expected in 2024.

xii. In July 2021, a staff member filed a complaint with the ILOAT against the Director-General's decision to maintain their initial performance qualification (fair). The Tribunal's ruling is expected in 2024.

xiii. In July 2021, a former staff member filed a complaint with the ILOAT against the Director-General's decision to refuse the recognition of total disability resulting in an incapacity to work. The Tribunal's ruling is expected in 2024.

xiv. In November 2021, a former staff member filed a complaint against the amount received from the insurer following an occupational accident. The Tribunal's ruling is expected in 2024.

xv. In January 2022, a staff member filed a complaint against the Director-General's decision to reject their internal appeal concerning an allegation of harassment. The Tribunal's ruling is expected in 2024.

xvi. In June 2022, two associated members of the personnel filed individual complaints with the ILOAT against the Director-General's decision to maintain the individual annual statements issued to them during the tax year 2021. The Tribunal's ruling is expected in 2025.

xvii. In July 2022, two staff members jointly filed a complaint with the ILOAT against the Director-General's decision to reject their internal appeals against their individual payslips, in the context of the 2021 five-yearly review. The Tribunal's ruling is expected in 2025.

xviii. In September 2022, two staff members individually filed complaints with the ILOAT against the Director-General's decision to declare their internal appeals receivable only to the extent that they challenged their performance qualification for 2021 (strong), in the context of the 2022 MERIT Guidelines. The Tribunal's ruling is expected in 2025.

xix. In September 2022, a CHIS member filed a complaint with the ILOAT against the Director-General's decision to reject the granting of an ex-gratia payment as per the CHIS Rules. The Tribunal's ruling is expected in 2025.

*[1] See Chapter VI of the Staff Rules and Regulations on "Settlement of Disputes and Discipline", also Operational Circular ("OC") 9 on "Principles and procedures governing complaints of harassment"; OC10 on "Principles and procedure governing investigation of fraud".*

*[2] The composition of the JAAB is described in Article R VI 1.10 of the Staff Regulations.*

*[3] The composition of the JADB is described in Article R VI 2.09 of the Staff Regulations.*

*[4] Items (i), (iv) and (v) of the requests for review, item (v) of the internal appeals and item (iii) of the complaints before the ILOAT concern the same MPE.*

*[5] Administrative Circular No. 14 (Rev. 5) on "Protection of members of the personnel against the financial consequences of illness, accident and incapacity for work".*

*[6] The predictions mentioned below as to the dates of the issuing of judgements by the ILOAT are based on the Organization's experience with the Tribunal. As the Tribunal may group cases together or deliver judgments in any order, more accurate estimates cannot be provided.*

*[7] For this reason, 54 of the original 59 complaints were withdrawn. The 55th complaint was rejected by the Tribunal as irreceivable.*

HR department

---

## CERN publishes its first F-Gas Policy

CERN has recently published a [Fluorinated Gases \(F-Gas\) Policy](#). This policy formalises CERN's commitment and strategy to minimise the impact on the environment of its installations containing and activities using F-Gases, thereby contributing to reducing its direct (Scope 1) greenhouse gas emissions.

[Greenhouse gas emissions at CERN](#) arise from the operation of the Laboratory's research facilities. The majority of emissions come from CERN's core experiments and more than 78% are a direct result

of the use of F-Gases. With climate change a growing concern, the Organization is engaged in reducing its direct greenhouse gas emissions, one of the key pillars of its [commitment to environmentally responsible research](#).

The policy will be implemented through a variety of measures ranging from an efficient management and monitoring of the procurement and use of F-gases to appropriate training of personnel concerned and proactive communication to stakeholders.

---

## Announcements

### Upcoming events

Check out this curated list of upcoming events for the CERN community

**3 Aug 14:00** | At CERN | R1 Terrace, Meyrin site |  
Music Club | EN & FR  
[Music on the Lawn](#)

**13 Aug 10:00** | Knowledge Sharing | CERN Library |  
Archives, Library and Open Science events | EN



### [Meet the authors of "New Challenges and Opportunities in Physics Education"](#)

**28 Aug 14:00** | Accelerators | Zoom | International Muon Collider Collaboration | EN  
[Early career researchers and muon colliders](#)

**10 Sep 17:30** | Knowledge Sharing | Council Chamber | Staff Association | FR  
[« Perdre le Nord », par Bernard Voyer](#)

**17 Sep 18:00** | At CERN | R1 lawn | CERN70 | EN & FR  
[CERN70 Community Event](#) (more [CERN70](#) events)

**Registration now open**

**22 Sep–5 Oct** | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN  
[Introduction to Accelerator Physics](#)

**11 Oct 14:00** | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 12 – Demystifying creativity](#)

**27 Nov–6 Dec** | Experiments | CERN | DRD1 | EN  
[DRD1 Gaseous Detectors School](#)

This is a curated list of events relevant to the CERN community. More events are available here: [home.cern/events](https://home.cern/events)

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact [bulletin-editors@cern.ch](mailto:bulletin-editors@cern.ch)

---

## REMINDER: Invitation to the CERN70 community event

In celebration of our Laboratory's 70th anniversary, we are thrilled to invite you to a special event. This occasion honours the scientific excellence, spirit of international collaboration, openness and inclusiveness that have driven our community to the forefront of scientific discovery and innovation for seven decades.

The celebration will take place on the Meyrin site on **Tuesday 17 September 2024**, from **6 p.m. to 10 p.m. CEST**, on the large lawn in front of Restaurant 1 and its surroundings.

All holders of a [valid CERN access card](#), including the family members of the personnel (spouse, registered partner, children), are warmly invited to join the event. Please note that visitor passes are not permitted on that day.

Prepare for an evening of diverse entertainment featuring musical performances, special guests

and many surprises. A variety of world street food and drinks will be available to everyone, free of charge.

[Register on Indico](#) (please log in to see the registration link on the menu bar on the left) at your earliest convenience but **no later than 31 August**. By registering, you will receive a ticket and QR code to **collect the food and drink vouchers at the indicated [days, times and locations](#)**. People who do not have a CERN account will be contacted via email shortly with instructions on how to register.

More practical information about site access, parking and shuttle services will be available in early September and communicated to all registered participants.

We look forward to celebrating this important milestone with you!

---

## REMINDER: CERN70 Community Event - Call for volunteers

**Sign up now to join the ranks of the volunteers for this special community event!**

We are looking for dedicated volunteers to help us ensure the event runs smoothly. Volunteers will be assigned to distribute access bracelets prior the event at lunchtime and, on the day, to info desks,

info patrols and to provide logistics assistance on and around the stage.

To ensure that volunteers can also enjoy the event, those volunteering on the 17 September will be

assigned to either the first half (17:00 to 20:15) or the second half (19:45 to 23:00) of the party.

By volunteering, you will be an active player and contribute to the success of this event.

**We hope that you will be interested and available to participate in this significant event for our community!**

**Sign up here:**

<https://indico.cern.ch/event/1427022/registrations/106635/>

---

## Winner of the CERN70 T-shirt design competition



*The official T-shirt featuring this winning design will be available to purchase at cost price during the CERN70 Community Event on 17 September. (Image: Helena Jaworska/CERN)*

Congratulations to Helena Jaworska, the winner of the [CERN70 Community Event T-shirt design competition](#). The official T-shirt featuring this winning design will be available to purchase at cost price during the [CERN70 Community Event on 17 September](#).

You have until 31 August to [register](#) for the CERN70 Community Event on 17 September – a celebration not to be missed! You can also [sign up to volunteer](#) to help make the event a success and peak behind-the-scenes – your help will be invaluable (and you'll be offered a specially branded volunteer T-shirt).

“Helena’s creative design captivated the jury and stood out for its visual quality, originality and the engaging way it captures the spirit of CERN,” explains Fabienne Landua, competition organiser and head of CERN’s Design and Visual identity team.

Thank you to everyone who participated and shared their talents. Submitted entries will be displayed on the Mezzanine in the Main Building from 31 July to 13 August.

---

## The new CERN Community Support Centre

**From 1 August, CERN’s former Main Reception area will be a hub for all CERN community services, including those previously available in Building 55**

The area in Building 33 that was formerly occupied by the Main Reception has undergone a complete transformation in order to house the new CERN Community Support Centre (CCSC), which will open on **1 August 2024**.

The CCSC will be operated by the CERN Service Desk team from **7.00 am to 6.30 pm, Monday to Friday**. It will be the first line of support for all

CERN services and activities that were previously housed in Building 55: contractor registration, deliver of access cards and professional visitor cards, biometry registration and dosimeter distribution, delivery of Swiss legitimization and French MAE cards, online training stations and general online and on-site assistance. These

services will all be centralised on the ground floor of Building 33.

After a complete renovation of the premises and an overhaul of procedures and tools, which focussed on digitalising arrival and departure formalities, the new CCSC will provide a modern and welcoming centre for members of the CERN community and professionals alike. From its glass roof to its flooring, the layout of the reception areas, the spatial flow, the furniture and the selection of artworks on display, every detail of the CCSC has been meticulously designed with the user in mind, offering a seamless and streamlined

experience for all visitors. The area that was previously occupied by the Main Reception has been redesigned by architects and fully renovated and rethought by the SCE teams, who worked in close collaboration with various departments (IT, HSE, HR, IR). This has resulted in a more functional, inclusive and user-friendly space geared towards providing a fluid user experience.

---

**For the latest news on the Service Portal and Service Now, read [their newsletter](#) or visit [their News page](#).**

---

## A Particular Score: Listen to cosmic rays at the new CERN Community Support Centre

**The installation by AATB, former artists in residence, transforms the constant flux of cosmic rays into an ever-evolving soundscape**

As visitors enter the new CERN Community Support Centre (CCSC), they will be surrounded by the cosmic soundscape created by *A Particular Score*, a newly commissioned sculpture by AATB. Located on the *Esplanade des Particules*, in the area previously occupied by CERN's Main Reception, the CCSC will offer comprehensive practical support to all members of the CERN community.

AATB, the artist group formed by Andrea Anner and Thibault Brevet, explores the idea of non-industrial robotics, experimenting with industrial automation processes and robotic arms. In 2023, AATB were in residence at CERN as the artists selected for Connect, the framework for collaboration between Arts at CERN and the Swiss Arts Council Pro Helvetia. Inspired by their residency, they redesigned their sonic sculpture *A Particular Score*, which will be on permanent display at the CCSC.

Cosmic rays are high-energy particles that originate from stars in the Milky Way and beyond. Although trillions of cosmic rays strike the Earth

every day, the atmosphere blocks most of them. The few that make it through create showers of secondary particles that reach the planet's surface, resulting in an invisible flux that continuously bathes our surroundings.

*A Particular Score* transforms this cosmic stream into a real-time score for visitors to the CCSC. The sculpture detects a particle and responds by striking one of the twelve quartz crystal tubes, creating an unpredictable sequence of audible tones. The soundscape evolves unexpectedly and randomly with each impact, giving us a glimpse of a phenomenon that surrounds us but is normally imperceptible.

*A Particular Score* was originally commissioned by Bienal da Maia, Portugal, in 2019, and was reworked for the Cosmos at *Grand Hornu* exhibition in Belgium in 2021. The present version was inspired by their Arts at CERN residency in 2023 and was shown at Mudac, Lausanne, Switzerland, as part of the *Space is the Place* exhibition held the same year.

Ana Prendes

---

## The Locks and Keys service is moving to Building 73

Please note that, as of 26 July 2024, the [Locks and Keys service](#) will be located in **Building 73**

(73/R-002), next to the CERN Stores. The opening hours will remain unchanged (**7.30 a.m. to 12.00**



noon and 1.00 p.m. to 4.30 p.m., Mondays to Fridays 7).

The service will continue to operate as before, with no change to the procedures for submitting requests.

Thank you for your understanding.

*SCE department*

---

## Ombud's Corner

### When disagreements or conflicts arise - don't let them get out of control

Disagreements are part of everyday life. They cannot be avoided and are sometimes necessary to bring about healthy change. Disagreements are the manifestation of tensions and of diverging needs, interests and values between people who are sometimes interdependent; this is especially true for disagreements between colleagues at work.

If, as a manager, you intervene early without trying to establish who is "right" and who is "wrong", you might be able to stop a simple disagreement escalating into a conflict. For this, you need to understand the factors underlying situations of tension at work.

Show the members of your team that you're available and that they should feel free to talk about the difficulties they're facing at work; if you set aside time for team discussions around working methods and conditions, the quality of work and the distribution of labour, you might stay ahead of the game and prevent disagreements from turning into full-blown conflicts.

Once a conflict has developed, don't imagine that there's a quick and easy fix. You should rather try to understand the root causes - how did it start and why? What were the points of disagreement that brought the parties into conflict and, conversely, what are the points of agreement?

Try to transform the conflict into something positive. If we hurl insults at one another, the conflict will have destructive consequences. But if we take the time to sit down in a peaceful atmosphere and look for a solution, we create the space needed to resolve the situation in a constructive way. Changing the nature of a conflict

means looking for the most reasonable way to resolve it.

Conflicts can sometimes be hidden from view and hard to identify, and those involved may prefer to say nothing and wait for the storm to blow over. On several occasions when people have come to me as the Ombud, they have admitted that they said nothing so as not to cause trouble, to keep the peace or out of fear of reactions. The risk is that the pent-up emotions, be they sadness or anger, will eventually come to the surface and cause hurt and damage.

It's in everyone's interest to settle differences without losing our temper. Calling on a trusted third party, either someone in our immediate work environment or through the support channels in place at CERN (of which the Ombud is one), can help avoid problems spiralling out of control.

Disagreements and conflicts are part of our everyday lives – the key is not to avoid them but to avoid escalation by acting early and by looking for help from a third party if the situation risks damaging working relations.

*Marie-Luce Falipou*

*I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at <https://mattermost.web.cern.ch/cern-ombud/>. The Ombud is available from Monday to Friday in office B500/1-004 on the Meyrin site. To make an appointment, in person or online, contact the Ombud at [ombuds@cern.ch](mailto:ombuds@cern.ch).*

*More information can be found on the Ombud's website: <https://ombuds.web.cern.ch/>*